

Haswell's 4.5W wonder

Intel has showcased a new processor for extremely thin tablets which may not require a fan to cool it at all, since it consumes a mere 4.5 Watts

Wi-Fi dead zone?

This Quora thread explains how download speeds drop to zero exactly above your Wi-Fi antenna. <http://dgit.in/DeadWiFi>

board we noticed a peculiarity whilst overclocking. The ECS board bundles an overclocking utility called eOC which wasn't really that helpful as it was just showing voltages that too of components outside the processor such as RAM, chipset and so on. We could overclock the processor using Intel's Extreme Tuning Utility. But despite setting all cores say 46x which gave a speed of 4600MHz, the processor speeds fell to 4400MHz while running the Cinebench R11.5. This was noted on all over-clocked frequencies.

Over 20k

This segment is for the niche users who know exactly what they want from their systems. Since all boards cost upwards of 20k, they are packed to the gills with features and are expected to come with a lot of bundled goodies.

Gigabyte G1 Sniper 5 has come to be associated with gaming just like the ROG for ASUS boards. The board comes in the traditional black/green colour tones with a beautifully designed heat sink which also offers active cooling.

The heatsink around the VRM is connected to the southbridge. Tube connectors

are also present on the heatsink to further help dissipate heat if you plan to use liquid cooling solutions. the G1 Sniper logo on the southbridge heat-sink looks impressive. The audio section which uses Creative audio processors, is the USP of the board as it is the first board to introduce an audio section which has removable operational amplifiers for analog headphones. Gigabyte also bundles in two Op-Amps along with the board but you can add your own. The audio section as expected, is isolated from the rest of the board. Dual gigabit LAN ports with one of them being controlled by the Killer E2200 processor is another feature targeted at LAN gamers. It also bundles in an 802.11n Wi-Fi and Bluetooth module which fits in one of the PCIe x1 slots.

Maximus VI Extreme is the flagship ROG board from ASUS and packs in too many features to list out in this limited space. There are five PCIe 3.0 x16 and you will need to refer to the manual to check which slots need to be used based on the number of cards you have. It has PCIe slot controllers present just behind the memory modules. So in case you do not want to use all the four graphics cards connected to your system, you can switch certain PCIe x16 slots off. There are voltage readout points just in front of the power/reset buttons. The innovative feature introduced with the Maximus VI Extreme is the ROG OC Panel (see box).

Of the other two ASUS boards the first is the Sabertooth TUF series board which comes with a 5-year warranty.



ASUS Z87 Sabertooth

The board has similar design elements from the previous generation which includes the thermal armour, brown coloured memory and PCIe slots. The armour has air flow valves around the VRM region. There are no buttons on the board apart from the MemOK button just behind the memory slots. There are three thermal sensor pins located just beside the USB connectors which are connected to thermistors. You can monitor temperatures of the CPU and GPU regions on the Thermal Radar utility. The board also bundles dust defenders which protect your PCIe and DRAM slots and two minute fans which can either be placed at the backpanel IO. The board also has a strengthening plate on the other side for giving more stability to the board and prevent it from bending.

The ASUS Z87-Deluxe is the flagship board meant for enthusiasts who do not want all the bells and whistles associated with an ROG board. It comes with an elaborate heat-sinks, ten SATA ports, on-board Wi-fi and Bluetooth adapter and three PCIe x16 slots. It supports 802.11ac standard which makes your system future proof to an extent as 802.11ac is the successor to 802.11n. It comes with the Dual Intelligent Processors IV moniker which allows you to tune your system for fan control, power control and energy efficiency. When you overclock the processor, the 4-way optimisa-

tion process optimises the fan speeds, energy efficiency, power ratings and runs on the fly stress tests to get a stable overclock.

Since this category of boards is meant for the enthusiast crowd overclocking was a breeze and we comfortably scaled the 4.7GHz mark on the Intel Core i7-4770K. The Maximus VI Extreme offered the most number of ways to overclock your system some of which may seem a bit round-about to some.

MicroATX boards

This is one segment which is quite interesting if seen from the perspective of system builders who are looking at a compact system. Not everyone has the luxury of space to house full tower cabinets, in such cases these boards with the microATX form factor cabinets are a good alternative. Of course for those looking at an even more compact build, always have the option to use miniITX boards – with or without an on-board processor. This time around we decided to have a separate category for these boards. We had two regular boards from ASRock and Gigabyte each, one gaming related and one high warranty one from ASUS.

ASRock Z87M Extreme 4 is a younger sibling of Z87 Extreme 4 and follows all the design philosophies seen on it. The heatsinks are quite good with golden accents and it has three PCIe x16 slots. The audio section is isolated from the rest of the board as is seen on its ATX cousins.

Gigabyte Z87M-D3H looks a bit similar to the ASRock Z87M Extreme 4 except for the golden coloured capacitors and more PCIe x16 slots and comparatively chunkier heatsinks



ASUS Maximus VI Extreme